SUPPLY CHAIN INTELLIGENCE // FREE CASE STUDY // SPACE // INDIA

NEWSPACE INDIA LIMITED (NSIL)

India's Commercial Space Gateway Has One Engine and It Belongs to Someone Else

Defense.Codes Case Study // Published Q2 2026 // Based on publicly available information only

NewSpace India Limited is the Government of India's commercial arm for space activities — the entity mandated to commercialise ISRO technology, manufacture launch vehicles, and anchor India's emerging private space sector. NSIL produces the GSLV Mk III (LVM3) — the rocket that launched OneWeb satellites and India's Chandrayaan-3. Its cryogenic upper stage engine — the CE-20 — is designed, qualified, and exclusively manufactured by ISRO's Liquid Propulsion Systems Centre in Thiruvananthapuram. NSIL is a commercial entity with a government monopoly supplier for its most critical component. No private backup. No alternate qualified source. No plan disclosed for engine supply chain independence. This is the chokepoint that limits India's commercial launch ambitions.

1	ZERO	LVM3	2047
CE-20 ENGINE MANUFACTURER GLOBALLY — ISRO LPSC THIRUVANANTHAPURAM	PRIVATE SECTOR QUALIFIED CRYOGENIC ENGINE MAKERS IN INDIA	NSIL'S PRIMARY LAUNCH VEHICLE — FULLY ISRO-ENGINE DEPENDENT	INDIA SPACE ECONOMY TARGET — ENGINE SUPPLY UNRESOLVED

Classification: FREE PUBLIC CASE STUDY // Defense.Codes

defense.codes | ceo@capa.cloud | +91 91637 62717

INTELLIGENCE ALERT // NSIL // THE CRYOGENIC ENGINE MONOPOLY

THE CORE FINDING

NewSpace India Limited is mandated to commercialise Indian space capability. It has successfully launched OneWeb satellites, signed contracts with international customers, and anchored India's aspirations to capture 10% of the global launch market by 2030. Every launch it performs on the LVM3 uses a CE-20 cryogenic engine that only ISRO can make. ISRO is not a commercial entity. ISRO has its own launch schedule priorities. ISRO's manufacturing capacity is government-funded, government-prioritised, and government-paced. NSIL's ability to meet commercial launch commitments is bounded by a government agency's production rate. That is not a commercial supply chain. That is a dependency masquerading as one.

THE CE-20 MONOPOLY // ONE ENGINE. ONE MAKER. NO BACKUP.

The CE-20 is a cryogenic engine that burns liquid hydrogen and liquid oxygen. It produces 19–20 tonnes of thrust and powers the C25 cryogenic upper stage of the LVM3. It is the most capable engine in India's launch vehicle inventory and the primary differentiator that enables NSIL to compete for GTO payloads and LEO constellation launches. It is manufactured exclusively at ISRO's Liquid Propulsion Systems Centre in Thiruvananthapuram, Kerala — a government R&D; and manufacturing facility whose primary mandate is ISRO mission support, not commercial production volume.

RISK DIMENSION	DETAIL	NSIL COMMERCIAL IMPACT	SEVERITY
Single manufacturer	ISRO LPSC Thiruvananthapuram — only qualified CE-20 production site	NSIL launch rate bounded by ISRO production schedule — cannot accelerate independently	CRITICAL
Government priority vs commercial schedule	ISRO's own missions (Gaganyaan, Chandrayaan-4, NISAR) take production priority	Commercial customer delivery dates at risk when ISRO mission schedules shift	CRITICAL
No private sector alternative	Agnikul, Skyroot, Bellatrix — all developing engines. None qualified for CE-20 class thrust (19–20T). Minimum 5–7 years.	NSIL has no fallback for LVM3 engine supply for at least 2030–2032	CRITICAL
Cryogenic supply chain	Liquid hydrogen and liquid oxygen supply. ISRO's IPRC Mahendragiri is primary test facility	Geographic concentration at LPSC and IPRC — both in South India	HIGH
Knowledge concentration	CE-20 design knowledge concentrated at LPSC — limited private sector technology transfer	Private sector cannot independently solve the supply problem — needs ISRO transfer	HIGH

THE COMMERCIAL CONSEQUENCE // ONEWEB WAS THE FIRST SIGNAL

NSIL successfully launched 36 OneWeb satellites per mission using LVM3 in 2022–2023. Those launches established NSIL as a credible international commercial launch provider. But every customer asking NSIL for a launch window commitment is asking NSIL to commit to a production schedule that NSIL does not control. NSIL can offer a date. ISRO controls whether the engine that date depends on is ready.

The competitive consequence: SpaceX Falcon 9 can commit to a launch window with high confidence because it manufactures its own engines at scale. Arianespace has ESA-origin engine supply with long-term production agreements. Rocket Lab manufactures Rutherford engines in-house at production scale. NSIL does not manufacture its own engines. Every commercial customer comparing NSIL to these alternatives is comparing a government-dependent supply chain to vertically integrated commercial ones.

THE PATH FORWARD

The solution is not complicated. India's IN-SPACe framework was designed to enable private sector participation in exactly this kind of technology transfer. The CE-20 design could be licensed to a qualified Indian private manufacturer under a DPSU framework. The timeline is 5–7 years. The cost is real but finite. The alternative — NSIL continuing to commercialise on an ISRO engine that ISRO controls — caps India's launch market ambitions at ISRO's production rate permanently.

WHAT A FULL ENGAGEMENT WOULD ADD

A full Defense.Codes engagement would add: (1) CE-20 production rate analysis — LPSC throughput vs NSIL commercial order pipeline; (2) Private sector engine qualification timeline model (Agnikul, Skyroot, Bellatrix); (3) IN-SPACe technology transfer framework assessment for CE-20 licensing; (4) NSIL commercial pipeline vs engine production gap model 2026–2032; (5) Competitor benchmark: SpaceX, Rocket Lab, Arianespace vertical integration vs NSIL.

This case study is one of five free public intelligence samples published by Defense.Codes. Category: SPACE — INDIA. Full-depth engagement reports available for purchase. Contact: ceo@capa.cloud | defense.codes

---

Prepared by Defense.Codes // defense.codes | ceo@capa.cloud | +91 91637 62717 // This is a free public case study demonstrating Defense.Codes intelligence methodology. Based entirely on publicly available information. Does not constitute investment advice, legal advice, or classified intelligence. All analysis is the independent assessment of Defense.Codes.

Sources: NSIL (newspaceindia.gov.in) corporate publications — ISRO annual reports 2024–2025 — IN-SPACe policy framework documents — OneWeb-NSIL launch contract announcements — Chandrayaan-3 mission documentation — GSLV Mk III / LVM3 technical specifications — India National Space Policy 2023 — NITI Aayog space economy report 2025 — Agnikul, Skyroot, Bellatrix public disclosures — DoS India budget documents.